전산화학 (2010년 1학기)

2차 과제 해답

(총 65 점)

1. **(총 25 점, 각 5 점)**

$$\mathbf{X} = \frac{m_a\mathbf{r}_a + m_b\mathbf{r}_b}{m_a + m_b}, \tag{1}$$

$$\mathbf{r} = \mathbf{r}_a - \mathbf{r}_b, \tag{2}$$

(a) (1) 식과 (2) 식으로부터 X 와 x 에 대해 다음과 같이 나타낼 수 있고,

$$X = \frac{m_a x_a + m_b x_b}{m_a + m_b}, \tag{3}$$

$$x = x_a - x_b, \tag{4}$$

일차 편도함수는 다음과 같이 얻어진다.

$$\left(\frac{\partial X}{\partial x_a}\right)_{x_b} = \frac{m_a}{m_a + m_b}, \tag{5}$$

$$\left(\frac{\partial x}{\partial x_a}\right)_{x_b} = 1, \tag{6}$$

$$\left(\frac{\partial X}{\partial x_b}\right)_{x_a} = \frac{m_b}{m_b + m_b}, \tag{7}$$

$$\left(\frac{\partial x}{\partial x_b}\right)_{x_a} = -1. \tag{8}$$

연쇄법칙을 이용하면 다음과 같은 결과를 얻는다.

$$\frac{\partial}{\partial x_a} = \frac{\partial X}{\partial x_a}\frac{\partial}{\partial X} + \frac{\partial x}{\partial x_a}\frac{\partial}{\partial x}, \tag{9}$$

$$= \frac{m_a}{m_a + m_b}\frac{\partial}{\partial X} + \frac{\partial}{\partial x}, \tag{10}$$

그리고,

$$\frac{\partial}{\partial x_b} = \frac{\partial X}{\partial x_b}\frac{\partial}{\partial X} + \frac{\partial x}{\partial x_b}\frac{\partial}{\partial x}, \tag{11}$$

$$= \frac{m_b}{m_b + m_b}\frac{\partial}{\partial X} - \frac{\partial}{\partial x}. \tag{12}$$

(b) 같은 방법으로, Y 와 y 에 대해 다음과 같이 나타낼 수 있고,

$$Y = \frac{m_a y_a + m_b y_b}{m_a + m_b}, \tag{13}$$

$$y = y_a - y_b, \tag{14}$$

일차 편도함수는 다음과 같이 얻어진다.

$$\left(\frac{\partial Y}{\partial y_a}\right)_{y_b} = \frac{m_a}{m_a+m_b}, \tag{15}$$

$$\left(\frac{\partial y}{\partial y_a}\right)_{y_b} = 1, \tag{16}$$

$$\left(\frac{\partial Y}{\partial y_b}\right)_{y_a} = \frac{m_b}{m_b+m_b}, \tag{17}$$

$$\left(\frac{\partial y}{\partial y_b}\right)_{y_a} = -1. \tag{18}$$

연쇄법칙을 이용하면 다음과 같은 결과를 얻는다.

$$\boxed{\begin{aligned}\frac{\partial}{\partial y_a} &= \frac{\partial Y}{\partial y_a}\frac{\partial}{\partial Y}+\frac{\partial y}{\partial y_a}\frac{\partial}{\partial y}, \\ &= \frac{m_a}{m_a+m_b}\frac{\partial}{\partial Y}+\frac{\partial}{\partial y}, \end{aligned}} \tag{19}$$

그리고

$$\boxed{\begin{aligned}\frac{\partial}{\partial y_b} &= \frac{\partial Y}{\partial y_b}\frac{\partial}{\partial Y}+\frac{\partial y}{\partial y_b}\frac{\partial}{\partial y}, \\ &= \frac{m_b}{m_b+m_b}\frac{\partial}{\partial Y}-\frac{\partial}{\partial y}. \end{aligned}} \tag{20}$$

(c) (10) 식으로부터,

$$\begin{aligned}\frac{\partial^2}{\partial x_a^2} &= \frac{\partial}{\partial x_a}\cdot\frac{\partial}{\partial x_a}, \\ &= \frac{\partial}{\partial x_a}\left(\frac{m_a}{m_a+m_b}\frac{\partial}{\partial X}+\frac{\partial}{\partial x}\right). \end{aligned} \tag{21}$$

g를 다음과 같이 정의한다면,

$$g \equiv \frac{m_a}{m_a+m_b}\frac{\partial}{\partial X}+\frac{\partial}{\partial x}, \tag{22}$$

(21) 식은 다음과 같이 쓸 수 있다.

$$\begin{aligned}\frac{\partial^2}{\partial x_a^2} &= \frac{\partial}{\partial x_a}\cdot\frac{\partial}{\partial x_a}, \\ &= \frac{\partial}{\partial x_a}\left(\frac{m_a}{m_a+m_b}\frac{\partial}{\partial X}+\frac{\partial}{\partial x}\right), \\ &= \frac{\partial g}{\partial x_a}, \\ &= \frac{\partial X}{\partial x_a}\frac{\partial g}{\partial X}+\frac{\partial x}{\partial x_a}\frac{\partial g}{\partial x}, \qquad \text{(from Eq. (9))} \\ &= \frac{m_a}{m_a+m_b}\frac{\partial g}{\partial X}+\frac{\partial g}{\partial x}, \qquad \text{(from Eqs. (5) and (6))} \\ &= \frac{m_a}{m_a+m_b}\frac{\partial}{\partial X}\left(\frac{m_a}{m_a+m_b}\frac{\partial}{\partial X}+\frac{\partial}{\partial x}\right)+\frac{\partial}{\partial x}\left(\frac{m_a}{m_a+m_b}\frac{\partial}{\partial X}+\frac{\partial}{\partial x}\right), \\ &= \boxed{\frac{m_a^2}{(m_a+m_b)^2}\frac{\partial^2}{\partial X^2}+\frac{2m_a}{m_a+m_b}\frac{\partial^2}{\partial X\partial x}+\frac{\partial^2}{\partial x^2}.} \end{aligned} \tag{23}$$

같은 방법으로 (12) 식으로부터,

$$\begin{aligned}\frac{\partial^2}{\partial x_b^2} &= \frac{\partial}{\partial x_b}\cdot\frac{\partial}{\partial x_b},\\ &= \frac{\partial}{\partial x_b}\left(\frac{m_b}{m_a+m_b}\frac{\partial}{\partial X}-\frac{\partial}{\partial x}\right).\end{aligned} \tag{24}$$

h를 다음과 같이 정의한다면,

$$h \equiv \frac{m_b}{m_a+m_b}\frac{\partial}{\partial X}-\frac{\partial}{\partial x}, \tag{25}$$

(24) 식은 다음과 같이 쓸 수 있다.

$$\begin{aligned}\frac{\partial^2}{\partial x_b^2} &= \frac{\partial}{\partial x_b}\cdot\frac{\partial}{\partial x_b},\\ &= \frac{\partial}{\partial x_b}\left(\frac{m_b}{m_a+m_b}\frac{\partial}{\partial X}-\frac{\partial}{\partial x}\right),\\ &= \frac{\partial h}{\partial x_b},\\ &= \frac{\partial X}{\partial x_b}\frac{\partial h}{\partial X}+\frac{\partial x}{\partial x_b}\frac{\partial h}{\partial x}, \qquad \text{(from Eq. (11))}\\ &= \frac{m_b}{m_a+m_b}\frac{\partial h}{\partial X}+\frac{\partial h}{\partial x}, \qquad \text{(from Eqs. (7) and (8))}\\ &= \frac{m_b}{m_a+m_b}\frac{\partial}{\partial X}\left(\frac{m_b}{m_a+m_b}\frac{\partial}{\partial X}-\frac{\partial}{\partial x}\right)+\frac{\partial}{\partial x}\left(\frac{m_b}{m_a+m_b}\frac{\partial}{\partial X}-\frac{\partial}{\partial x}\right),\\ &= \boxed{\frac{m_b^2}{(m_a+m_b)^2}\frac{\partial^2}{\partial X^2}-\frac{2m_b}{m_a+m_b}\frac{\partial^2}{\partial X\partial x}+\frac{\partial^2}{\partial x^2}.}\end{aligned} \tag{26}$$

(d) (c)와 정확히 같은 방식으로,

$$\boxed{\frac{\partial^2}{\partial y_a^2} = \frac{m_a^2}{(m_a+m_b)^2}\frac{\partial^2}{\partial Y^2}+\frac{2m_a}{m_a+m_b}\frac{\partial^2}{\partial Y\partial y}+\frac{\partial^2}{\partial y^2},} \tag{27}$$

그리고

$$\boxed{\frac{\partial^2}{\partial y_b^2} = \frac{m_b^2}{(m_a+m_b)^2}\frac{\partial^2}{\partial Y^2}-\frac{2m_b}{m_a+m_b}\frac{\partial^2}{\partial Y\partial y}+\frac{\partial^2}{\partial y^2}.} \tag{28}$$

(e) (23) 식, (26) 식, (27) 식, 그리고 (28) 식으로부터,

$$\begin{aligned}
\frac{1}{m_a}\nabla_a^2+\frac{1}{m_b}\nabla_b^2 &= \frac{1}{m_a}\left(\frac{\partial^2}{\partial x_a^2}+\frac{\partial^2}{\partial y_a^2}\right)+\frac{1}{m_b}\left(\frac{\partial^2}{\partial x_b^2}+\frac{\partial^2}{\partial y_b^2}\right), \qquad (29)\\
&= \frac{1}{m_a}\left(\frac{m_a^2}{(m_a+m_b)^2}\frac{\partial^2}{\partial X^2}+\frac{2m_a}{m_a+m_b}\frac{\partial^2}{\partial X\partial x}+\frac{\partial^2}{\partial x^2}\right.\\
&\quad \left.+\frac{m_a^2}{(m_a+m_b)^2}\frac{\partial^2}{\partial Y^2}+\frac{2m_a}{m_a+m_b}\frac{\partial^2}{\partial Y\partial y}+\frac{\partial^2}{\partial y^2}\right)\\
&\quad +\frac{1}{m_b}\left(\frac{m_b^2}{(m_a+m_b)^2}\frac{\partial^2}{\partial X^2}-\frac{2m_b}{m_a+m_b}\frac{\partial^2}{\partial X\partial x}+\frac{\partial^2}{\partial x^2}\right.\\
&\quad \left.+\frac{m_b^2}{(m_b+m_b)^2}\frac{\partial^2}{\partial Y^2}-\frac{2m_b}{m_b+m_b}\frac{\partial^2}{\partial Y\partial y}+\frac{\partial^2}{\partial y^2}\right),\\
&= \frac{m_a}{(m_a+m_b)^2}\frac{\partial^2}{\partial X^2}+\frac{2}{m_a+m_b}\frac{\partial^2}{\partial X\partial x}+\frac{1}{m_a}\frac{\partial^2}{\partial x^2}\\
&\quad +\frac{m_a}{(m_a+m_b)^2}\frac{\partial^2}{\partial Y^2}+\frac{2}{m_a+m_b}\frac{\partial^2}{\partial Y\partial y}+\frac{1}{m_a}\frac{\partial^2}{\partial y^2}\\
&\quad +\frac{m_b}{(m_a+m_b)^2}\frac{\partial^2}{\partial X^2}-\frac{2}{m_a+m_b}\frac{\partial^2}{\partial X\partial x}+\frac{1}{m_b}\frac{\partial^2}{\partial x^2}\\
&\quad +\frac{m_b}{(m_b+m_b)^2}\frac{\partial^2}{\partial Y^2}-\frac{2}{m_a+m_b}\frac{\partial^2}{\partial Y\partial y}+\frac{1}{m_b}\frac{\partial^2}{\partial y^2},\\
&= \frac{m_a}{(m_a+m_b)^2}\left(\frac{\partial^2}{\partial X^2}+\frac{\partial^2}{\partial Y^2}\right)+\frac{m_b}{(m_a+m_b)^2}\left(\frac{\partial^2}{\partial X^2}+\frac{\partial^2}{\partial Y^2}\right)\\
&\quad +\frac{1}{m_a}\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right)+\frac{1}{m_b}\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right),\\
&= \boxed{\frac{1}{m_a+m_b}\left(\frac{\partial^2}{\partial X^2}+\frac{\partial^2}{\partial Y^2}\right)+\left(\frac{1}{m_a}+\frac{1}{m_b}\right)\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right).} \qquad (30)
\end{aligned}$$

2. **(총 15 점, 각 5 점)**

$$x=r\cos\phi,\quad y=r\sin\phi\Rightarrow r^2=x^2+y^2,\quad \tan\phi=\frac{y}{x}. \qquad (31)$$

(a) 연쇄법칙을 이용하면,

$$\left(\frac{\partial}{\partial x}\right)_y = \left(\frac{\partial r}{\partial x}\right)_y\left(\frac{\partial}{\partial r}\right)_\phi+\left(\frac{\partial \phi}{\partial x}\right)_y\left(\frac{\partial}{\partial \phi}\right)_r, \qquad (32a)$$

$$\left(\frac{\partial}{\partial y}\right)_x = \left(\frac{\partial r}{\partial y}\right)_x\left(\frac{\partial}{\partial r}\right)_\phi+\left(\frac{\partial \phi}{\partial y}\right)_x\left(\frac{\partial}{\partial \phi}\right)_r, \qquad (32b)$$

이고,

$$\frac{\partial r}{\partial x}=\frac{x}{r}=\cos\phi,\quad \frac{\partial\phi}{\partial x}=-\frac{\sin\phi}{r}, \qquad (33a)$$

$$\frac{\partial r}{\partial y}=\frac{y}{r}=\sin\phi,\quad \frac{\partial\phi}{\partial y}=\frac{\cos\phi}{r}, \qquad (33b)$$

이므로,

$$\boxed{\begin{aligned} \frac{\partial}{\partial x} &= \cos\phi\frac{\partial}{\partial r} - \frac{\sin\phi}{r}\frac{\partial}{\partial\phi}, \quad \text{(34a)} \\ \frac{\partial}{\partial y} &= \sin\phi\frac{\partial}{\partial r} + \frac{\cos\phi}{r}\frac{\partial}{\partial\phi}. \quad \text{(34b)} \end{aligned}}$$

(b) g를 다음과 같이 정의하면,

$$g \equiv \frac{\partial}{\partial x} = \cos\phi\frac{\partial}{\partial r} - \frac{\sin\phi}{r}\frac{\partial}{\partial\phi}, \tag{35}$$

$\partial^2/\partial x^2$는 다음과 같이 쓸 수 있다.

$$\begin{aligned} \frac{\partial^2}{\partial x^2} &= \frac{\partial}{\partial x}\cdot\frac{\partial}{\partial x}, \\ &= \frac{\partial g}{\partial x}, \\ &= \frac{\partial r}{\partial x}\frac{\partial g}{\partial r} + \frac{\partial\phi}{\partial x}\frac{\partial g}{\partial\phi}, \qquad (\text{(32a) 식으로부터}) \\ &= \cos\phi\frac{\partial g}{\partial r} - \frac{\sin\phi}{r}\frac{\partial g}{\partial\phi}, \qquad (\text{(33a) 식과 (33b) 식으로부터}) \\ &= \cos\phi\frac{\partial}{\partial r}\left(\cos\phi\frac{\partial}{\partial r} - \frac{\sin\phi}{r}\frac{\partial}{\partial\phi}\right) - \frac{\sin\phi}{r}\frac{\partial}{\partial\phi}\left(\cos\phi\frac{\partial}{\partial r} - \frac{\sin\phi}{r}\frac{\partial}{\partial\phi}\right), \\ &= \cos^2\phi\frac{\partial^2}{\partial r^2} + \frac{\sin\phi\cos\phi}{r^2}\frac{\partial}{\partial\phi} - \frac{\sin\phi\cos\phi}{r}\frac{\partial^2}{\partial r\partial\phi} \\ &\quad + \frac{\sin^2\phi}{r}\frac{\partial}{\partial r} - \frac{\sin\phi\cos\phi}{r}\frac{\partial^2}{\partial\phi\partial r} + \frac{\sin\phi\cos\phi}{r^2}\frac{\partial}{\partial\phi} + \frac{\sin^2\phi}{r^2}\frac{\partial^2}{\partial\phi^2}. \\ &= \boxed{\cos^2\phi\frac{\partial^2}{\partial r^2} + \frac{2\sin\phi\cos\phi}{r^2}\frac{\partial}{\partial\phi} - \frac{2\sin\phi\cos\phi}{r}\frac{\partial^2}{\partial r\partial\phi} + \frac{\sin^2\phi}{r}\frac{\partial}{\partial r} + \frac{\sin^2\phi}{r^2}\frac{\partial^2}{\partial\phi^2}.} \end{aligned} \tag{36}$$

같은 방식으로 h를 다음과 같이 정의하면

$$h \equiv \frac{\partial}{\partial y} = \sin\phi\frac{\partial}{\partial r} + \frac{\cos\phi}{r}\frac{\partial}{\partial\phi}, \tag{37}$$

$\partial^2/\partial y^2$ 는 다음과 같이 쓸 수 있다.

$$\begin{aligned}
\frac{\partial^2}{\partial y^2} &= \frac{\partial}{\partial y}\cdot\frac{\partial}{\partial y}, \\
&= \frac{\partial h}{\partial y}, \\
&= \frac{\partial r}{\partial y}\frac{\partial h}{\partial r}+\frac{\partial \phi}{\partial y}\frac{\partial h}{\partial \phi}, \qquad \text{(Eq. (32b) 식으로부터)} \\
&= \sin\phi\frac{\partial h}{\partial r}+\frac{\cos\phi}{r}\frac{\partial h}{\partial \phi}, \qquad \text{((33a) 식과 (33b) 식으로부터)} \\
&= \sin\phi\frac{\partial}{\partial r}\left(\sin\phi\frac{\partial}{\partial r}+\frac{\cos\phi}{r}\frac{\partial}{\partial\phi}\right)+\frac{\cos\phi}{r}\frac{\partial}{\partial\phi}\left(\sin\phi\frac{\partial}{\partial r}+\frac{\cos\phi}{r}\frac{\partial}{\partial\phi}\right), \\
&= \sin^2\phi\frac{\partial^2}{\partial r^2}-\frac{\sin\phi\cos\phi}{r^2}\frac{\partial}{\partial\phi}+\frac{\cos\phi\sin\phi}{r}\frac{\partial^2}{\partial r\partial\phi} \\
&\quad +\frac{\cos^2\phi}{r}\frac{\partial}{\partial r}+\frac{\cos\phi\sin\phi}{r}\frac{\partial^2}{\partial\phi\partial r}-\frac{\cos\phi\sin\phi}{r^2}\frac{\partial}{\partial\phi}+\frac{\cos^2\phi}{r^2}\frac{\partial^2}{\partial\phi^2}. \\
&= \boxed{\sin^2\phi\frac{\partial^2}{\partial r^2}-\frac{2\cos\phi\sin\phi}{r^2}\frac{\partial}{\partial\phi}+\frac{2\cos\phi\sin\phi}{r}\frac{\partial^2}{\partial r\partial\phi}+\frac{\cos^2\phi}{r}\frac{\partial}{\partial r}+\frac{\cos^2\phi}{r^2}\frac{\partial^2}{\partial\phi^2}.}
\end{aligned} \tag{38}$$

(c) (36) 식과 (38) 식으로부터, 다음과 같은 결과를 얻는다.

$$\begin{aligned}
\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2} &= \cos^2\phi\frac{\partial^2}{\partial r^2}+\frac{2\sin\phi\cos\phi}{r^2}\frac{\partial}{\partial\phi}-\frac{2\sin\phi\cos\phi}{r}\frac{\partial^2}{\partial r\partial\phi}+\frac{\sin^2\phi}{r}\frac{\partial}{\partial r}+\frac{\sin^2\phi}{r^2}\frac{\partial^2}{\partial\phi^2} \\
&\quad +\sin^2\phi\frac{\partial^2}{\partial r^2}-\frac{2\cos\phi\sin\phi}{r^2}\frac{\partial}{\partial\phi}+\frac{2\cos\phi\sin\phi}{r}\frac{\partial^2}{\partial r\partial\phi}+\frac{\cos^2\phi}{r}\frac{\partial}{\partial r}+\frac{\cos^2\phi}{r^2}\frac{\partial^2}{\partial\phi^2}, \\
&= (\sin^2\phi+\cos^2\phi)\frac{\partial^2}{\partial r^2}+\frac{(\sin^2\phi+\cos^2\phi)}{r}\frac{\partial}{\partial r}+\frac{(\sin^2\phi+\cos^2\phi)}{r^2}\frac{\partial^2}{\partial\phi^2}, \\
&= \boxed{\frac{\partial^2}{\partial r^2}+\frac{\partial}{r\partial r}+\frac{\partial^2}{r^2\partial\phi^2}.}
\end{aligned} \tag{39}$$

3. **(10 점)** 삼차원 직교좌표와 구면 극좌표의 관계식으로부터

$$x = r\sin\theta\cos\phi, \tag{40a}$$
$$y = r\sin\theta\sin\phi, \tag{40b}$$
$$z = r\cos\theta, \tag{40c}$$

그리고

$$r^2 = x^2+y^2+z^2, \tag{41a}$$
$$\cos\theta = \frac{z}{\sqrt{x^2+y^2+z^2}}, \tag{41b}$$
$$\tan\phi = \frac{y}{x}. \tag{41c}$$

$\partial r/\partial x$, $\partial\theta/\partial x$, 그리고 $\partial\phi/\partial x$ 는 다음과 같이 구해진다.

(1) (41a) 식으로부터,

$$\frac{\partial r}{\partial x} = \frac{x}{r} = \sin\theta\cos\phi, \tag{42}$$

(2) (41b) 식으로부터,

$$\begin{aligned}\cos\theta &= \frac{z}{\sqrt{x^2+y^2+z^2}} \Rightarrow -\sin\theta d\theta = \frac{-xz}{(x^2+y^2+z^2)^{3/2}}dx = \frac{-xz}{r^3}dx, \\ &\Rightarrow \frac{\partial\theta}{\partial x} = \frac{xz}{r^3\sin\theta} = \frac{r\sin\theta\cos\phi r\cos\theta}{r^3\sin\theta} = \frac{\cos\theta\cos\phi}{r} \quad ((40a)\text{ 식과 }(40c)\text{ 식으로부터}).\end{aligned} \tag{43}$$

(3) Eq. (41c) 식으로부터,

$$\begin{aligned}\tan\phi &= \frac{y}{x} \Rightarrow \frac{1}{\cos^2\phi}d\phi = -\frac{y}{x^2}dx \\ &\Rightarrow \frac{\partial\phi}{\partial x} = -\cos^2\phi\frac{y}{x^2} = -\cos^2\phi\frac{r\sin\theta\sin\phi}{r^2\sin^2\theta\cos^2\phi} = -\frac{\sin\phi}{r\sin\theta}. \quad ((40a)\text{ 식과 }(40b)\text{ 식으로부터})\end{aligned} \tag{44}$$

연쇄법칙을 이용하면 $\partial/\partial x$는 다음과 같이 쓸 수 있다.

$$\boxed{\begin{aligned}\frac{\partial}{\partial x} &= \frac{\partial r}{\partial x}\frac{\partial}{\partial r} + \frac{\partial\theta}{\partial x}\frac{\partial}{\partial\theta} + \frac{\partial\phi}{\partial x}\frac{\partial}{\partial\phi}, \\ &= \sin\theta\cos\phi\frac{\partial}{\partial r} + \frac{\cos\theta\cos\phi}{r}\frac{\partial}{\partial\theta} - \frac{\sin\phi}{r\sin\theta}\frac{\partial}{\partial\phi}.\end{aligned}} \tag{45}$$

같은 방식으로

$$\frac{\partial r}{\partial y} = \frac{y}{r} = \sin\theta\sin\phi, \tag{46}$$

$$\begin{aligned}\cos\theta &= \frac{z}{\sqrt{y^2+y^2+z^2}} \Rightarrow -\sin\theta d\theta = \frac{-yz}{(y^2+y^2+z^2)^{3/2}}dy = \frac{-yz}{r^3}dy, \\ &\Rightarrow \frac{\partial\theta}{\partial y} = \frac{yz}{r^3\sin\theta} = \frac{r\sin\theta\sin\phi r\cos\theta}{r^3\sin\theta} = \frac{\cos\theta\sin\phi}{r} \quad (\text{from Eqs. (40b) and (40c)})\end{aligned} \tag{47}$$

그리고

$$\begin{aligned}\tan\phi &= \frac{y}{x} \Rightarrow \frac{1}{\cos^2\phi}d\phi = \frac{1}{x}dy \\ &\Rightarrow \frac{\partial\phi}{\partial y} = \cos^2\phi\frac{1}{x} = \cos^2\phi\frac{1}{r\sin\theta\cos\phi} = \frac{\cos\phi}{r\sin\theta}. \quad (\text{from Eqs. (40a) and (40b)})\end{aligned} \tag{48}$$

연쇄법칙을 이용하면, $\partial/\partial y$는 다음과 같이 얻어진다.

$$\boxed{\begin{aligned}\frac{\partial}{\partial y} &= \frac{\partial r}{\partial y}\frac{\partial}{\partial r} + \frac{\partial\theta}{\partial y}\frac{\partial}{\partial\theta} + \frac{\partial\phi}{\partial y}\frac{\partial}{\partial\phi}, \\ &= \sin\theta\sin\phi\frac{\partial}{\partial r} + \frac{\cos\theta\sin\phi}{r}\frac{\partial}{\partial\theta} + \frac{\cos\phi}{r\sin\theta}\frac{\partial}{\partial\phi}.\end{aligned}} \tag{49}$$

마지막으로 (40a), (40b), (45), (49) 식들을 이용하면, 다음의 결과를 얻을 수 있다.

$$\begin{aligned} x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x} &= r\sin\theta\cos\phi\left(\sin\theta\sin\phi\frac{\partial}{\partial r} + \frac{\cos\theta\sin\phi}{r}\frac{\partial}{\partial\theta} + \frac{\cos\phi}{r\sin\theta}\frac{\partial}{\partial\phi}\right) \\ &\quad -r\sin\theta\sin\phi\left(\sin\theta\cos\phi\frac{\partial}{\partial r} + \frac{\cos\theta\cos\phi}{r}\frac{\partial}{\partial\theta} - \frac{\sin\phi}{r\sin\theta}\frac{\partial}{\partial\phi}\right) \\ &= (\sin^2\phi + \cos^2\phi)\frac{\partial}{\partial\phi}, \\ &= \frac{\partial}{\partial\phi}. \end{aligned} \tag{50}$$

4. **(총 15 점, 각 5 점)**

(a) 온도와 부피에 대한 $S(T,V)$ 의 전미분은 다음과 같이 구할 수 있다.

$$dS = \left(\frac{\partial S}{\partial T}\right)_V dT + \left(\frac{\partial S}{\partial V}\right)_T dV. \tag{51}$$

(b) (51) 식의 양변을 압력 P 를 고정시키고 T 로 편미분을 하면 다음과 같은 식을 얻을 수 있다.

$$\left(\frac{\partial S}{\partial T}\right)_P = \left(\frac{\partial S}{\partial T}\right)_V + \left(\frac{\partial S}{\partial V}\right)_T\left(\frac{\partial V}{\partial T}\right)_P. \tag{52}$$

일정부피 열용량과 일정압력 열용량의 정의가 다음과 같이 주어지므로:

$$C_V = \frac{\delta q_V}{dT} = T\left(\frac{\partial S}{\partial T}\right)_V, \tag{53}$$

$$C_P = \frac{\delta q_P}{dT} = T\left(\frac{\partial S}{\partial T}\right)_P, \tag{54}$$

(53) 식과 (54) 식을 (52) 식에 대입하면, 다음의 결과를 얻는다.

$$\frac{C_P}{T} = \frac{C_V}{T} + \left(\frac{\partial S}{\partial V}\right)_T\left(\frac{\partial V}{\partial T}\right)_P \tag{55}$$

(c) $d\bar{A} = -SdT - PdV$ 에 대한 Maxwell 관계식:

$$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial P}{\partial T}\right)_V \tag{56}$$

과 순환 법칙:

$$\left(\frac{\partial P}{\partial T}\right)_V\left(\frac{\partial T}{\partial V}\right)_P\left(\frac{\partial V}{\partial P}\right)_T = -1 \Rightarrow \left(\frac{\partial P}{\partial T}\right)_V = -\left(\frac{\partial V}{\partial T}\right)_P\left(\frac{\partial P}{\partial V}\right)_T \tag{57}$$

을 (55) 식에 적용하면 다음과 같은 식을 얻을 수 있다.

$$C_P - C_V = -T\left(\frac{\partial P}{\partial V}\right)_T\left(\frac{\partial V}{\partial T}\right)_P^2 \tag{58}$$

참고: 등압 팽창 계수 (thermal expansion coefficient) 와 등온 압축률 (isothermal

compressibility)의 정의:

$$\beta = \frac{1}{V}\Big(\frac{\partial V}{\partial T}\Big)_P, \qquad \kappa = -\frac{1}{V}\Big(\frac{\partial V}{\partial P}\Big)_T \tag{59}$$

를 (58) 식에 적용하면, 다음과 같은 일정부피 열용량과 일정압력 열용량 사이의 관계식을 얻을 수 있다.

$$\boxed{C_P - C_V = \frac{TV\beta^2}{\kappa}} \tag{60}$$